

Syllabus of UNDERGRADUATE DEGREE COURSE

Artificial Intelligence and Data Science



Rajasthan Technical University, Kota
Effective from session: 2025 – 2026



RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Syllabus

B.Tech.: Artificial Intelligence and Data Science

5AID3-01: Data Mining-Concepts and Techniques

Credit: 2
2L+0T+0P

Max. Marks: 100(IA:30, ETE:70)
End Term Exam: 3 Hours

Course Objectives:

1. To introduce the fundamental processes data warehousing and major issues in data mining
2. To impart the knowledge on various data mining concepts and techniques that can be applied to text mining, web mining etc.
3. To develop the knowledge for application of data mining and social impacts of data mining.

Course Outcomes: After completion of the course, students would be able to:

1. Interpret the contribution of data warehousing and data mining to the decision-support systems.
2. Prepare the data needed for data mining using preprocessing techniques.
3. Extract useful information from the labeled data using various classifiers.
4. Compile unlabeled data into clusters applying various clustering algorithms.
5. Discover interesting patterns from large amounts of data using Association Rule Mining
6. Demonstrate capacity to perform a self-directed piece of practical work that requires the application of data mining techniques.

Detailed Syllabus: (per session plan)

UNIT	Contents
1	Introduction to Data Mining: Introduction to data mining-Data mining functionalities-Steps in data mining process- Classification of data mining systems, Major issues in data mining. Data Wrangling and Preprocessing: Data Preprocessing: An overview-Data cleaning-Data transformation and Data discretization
2	Predictive Modeling: General approach to classification-Decision tree induction- Bayes classification methods- advanced classification methods: Bayesian belief networks- Classification by Backpropagation- Support Vector Machines-Lazy learners
3	Descriptive Modeling: Types of data in cluster analysis-Partitioning methods- Hierarchical methods-Advanced cluster analysis: Probabilistic model-based clustering- Clustering high-dimensional data-Outlier analysis
4	Discovering Patterns and Rules: Frequent Pattern Mining: Basic Concepts and a Road Map - Efficient and scalable frequent item set mining methods: Apriori algorithm, FP-Growth algorithm- Mining frequent itemsets using vertical data format- Mining closed and max patterns- Advanced Pattern Mining: Pattern Mining in Multilevel, Multidimensional Space
5	Data Mining Trends and Research Frontiers: Other methodologies of data mining: Web mining- Temporal mining-Spatial mining-Statistical data mining- Visual and audio data mining- Data mining applications- Data mining and society: Ubiquitous and invisible data mining- Privacy, Security, and Social Impacts of data mining

TEXT BOOKS:

1. Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques, Morgan Kaufmann Publishers, third edition ,2013
2. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar, Introduction to Data Mining, second edition, Pearson, 2019

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B.Tech.: Artificial Intelligence and Data Science

REFERENCE BOOKS:

1. Ian.H.Witten, Eibe Frank and Mark.A.Hall, Data Mining:Practical Machine Learning Tools and Techniques,third edition , 2017
2. Alex Berson and Stephen J. Smith, Data Warehousing, Data Mining & OLAP, Tata McGraw Hill Edition, Tenth Reprint, 2008.
3. Hand, D., Mannila, H. and Smyth, P. Principles of Data Mining, MIT Press: Massachusets. third edition, Pearson, 2013



RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Syllabus

B.Tech.: Artificial Intelligence and Data Science

5AID4-02: Compiler Design

Credit: 3

Max. Marks: 100(IA:30, ETE:70)

3L+0T+0P

End Term Exam: 3 Hours

SN	Contents	Hours
1	Introduction: Objective, scope and outcome of the course.	01
2	Introduction: Objective, scope and outcome of the course. Compiler, Translator, Interpreter definition, Phase of compiler, Bootstrapping, Review of Finite automata lexical analyzer, Input, Recognition of tokens, Idea about LEX: A lexical analyzer generator, Error handling.	06
3	Review of CFG Ambiguity of grammars: Introduction to parsing. Top down parsing, LL grammars & parsers error handling of LL parser, Recursive descent parsing predictive parsers, Bottom up parsing, Shift reduce parsing, LR parsers, Construction of SLR, Conical LR & LALR parsing tables, parsing with ambiguous grammar. Operator precedence parsing, Introduction of automatic parser generator: YACC error handling in LR parsers.	10
4	Syntax directed definitions; Construction of syntax trees, S-Attributed Definition, L-attributed definitions, Top down translation. Intermediate code forms using postfix notation, DAG, Three address code, TAC for various control structures, Representing TAC using triples and quadruples, Boolean expression and control structures.	10
5	Storage organization; Storage allocation, Strategies, Activation records, Accessing local and non-local names in a block structured language, Parameters passing, Symbol table organization, Data structures used in symbol tables.	08
6	Definition of basic block control flow graphs; DAG representation of basic block, Advantages of DAG, Sources of optimization, Loop optimization, Idea about global data flow analysis, Loop invariant computation, Peephole optimization, Issues in design of code generator, A simple code generator, Code generation from DAG.	07
	Total	42



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B.Tech.: Artificial Intelligence and Data Science

5AID4-03: Operating System

Credit: 3

Max. Marks: 100(IA:30, ETE:70)

3L+0T+0P

End Term Exam: 3 Hours

SN	Contents	Hours
1	Introduction: Objective, scope and outcome of the course.	01
2	Introduction and History of Operating systems: Structure and operations; processes and files Processor management: inter process communication, mutual exclusion, semaphores, wait and signal procedures, process scheduling and algorithms, critical sections, threads, multithreading	04
3	Memory management: contiguous memory allocation, virtual memory, paging, page table structure, demand paging, page replacement policies, thrashing, segmentation, case study	05
4	Deadlock: Shared resources, resource allocation and scheduling, resource graph models, deadlock detection, deadlock avoidance, deadlock prevention algorithms Device management: devices and their characteristics, device drivers, device handling, disk scheduling algorithms and policies	15
5	File management: file concept, types and structures, directory structure, cases studies, access methods and matrices, file security, user authentication	07
6	UNIX and Linux operating systems as case studies; Time OS and case studies of Mobile OS	08
	Total	40



RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Syllabus

B.Tech.: Artificial Intelligence and Data Science

5AID4-04: Computer Graphics & Multimedia

Credit: 3
3L+0T+0P

Max. Marks: 100(IA:30, ETE:70)
End Term Exam: 3 Hours

SN	Contents	Hours
1	Introduction: Objective, scope and outcome of the course.	01
2	Basic of Computer Graphics: Basic of Computer Graphics, Applications of computer graphics, Display devices, Random and Raster scan systems, Graphics input devices, Graphics software and standards	06
3	Graphics Primitives: Points, lines, circles and ellipses as primitives, scan conversion algorithms for primitives, Fill area primitives including scan-line polygon filling, inside-outside test, boundary and flood-fill, character generation, line attributes, area-fill attributes, character attributes. Aliasing, and introduction to Anti Aliasing (No anti aliasing algorithm).	07
4	Two Dimensional Graphics: Transformations (translation, rotation, scaling), matrix representation, homogeneous coordinates, composite transformations, reflection and shearing, viewing pipeline and coordinates system, window-to-viewport transformation, clipping including point clipping, line clipping (cohen-sutherland, liang- bersky, NLN), polygon clipping	08
5	Three Dimensional Graphics: 3D display methods, polygon surfaces, tables, equations, meshes, curved lies and surfaces, quadric surfaces, spline representation, cubic spline interpolation methods, Bazier curves and surfaces, B-spline curves and surfaces. 3D scaling, rotation and translation, composite transformation, viewing pipeline and coordinates, parallel and perspective transformation, view volume and general (parallel and perspective) projection transformations.	08
6	Illumination and Colour Models: Light sources – basic illumination models – halftone patterns and dithering techniques; Properties of light – Standard primaries and chromaticity diagram; Intuitive colour concepts – RGB colour model – YIQ colour model – CMY colour model – HSV colour model – HLS colour model; Colour selection.	06
7	Animations & Realism: Design of Animation sequences – animation function – raster animation – key frame systems – motion specification – morphing – tweening. Computer Graphics Realism: Tiling the plane – Recursively defined curves – Koch curves – C curves – Dragons – space filling curves – fractals – Grammar based models – fractals – turtle graphics – ray tracing.	06
	Total	42



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Syllabus

B.Tech.: Artificial Intelligence and Data Science

5AID4-05: Analysis of Algorithms

Credit: 3
3L+0T+0P

Max. Marks: 100(IA:30, ETE:70)
End Term Exam: 3 Hours

SN	Contents	Hours
1	Introduction: Objective, scope and outcome of the course.	01
2	Background: Review of Algorithm, Complexity Order Notations: definitions and calculating complexity. Divide And Conquer Method: Binary Search, Merge Sort, Quick sort and Strassen's matrix multiplication algorithms.	06
3	Greedy Method: Knapsack Problem, Job Sequencing, Optimal Merge Patterns and Minimal Spanning Trees. Dynamic Programming: Matrix Chain Multiplication. Longest CommonSubsequence and 0/1 Knapsack Problem.	10
4	Branch And Bound: Traveling Salesman Problem and Lower Bound Theory. Backtracking Algorithms and queens problem. Pattern Matching Algorithms: Naïve and Rabin Karp string matching algorithms, KMP Matcher and Boyer Moore Algorithms.	08
5	Assignment Problems: Formulation of Assignment and Quadratic Assignment Problem. Randomized Algorithms- Las Vegas algorithms, Monte Carlo algorithms, randomized algorithm for Min-Cut, randomized algorithm for 2- SAT. Problem definition of Multicommodity flow, Flow shop scheduling and Network capacity assignment problems.	08
6	Problem Classes Np, Np-Hard And Np-Complete: Definitions of P, NP-Hard and NP-Complete Problems. Decision Problems.Cook's Theorem. Proving NP-Complete Problems - Satisfiability problem and Vertex Cover Problem. Approximation Algorithms for Vertex Cover andSet Cover Problem.	08
	Total	41

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RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Syllabus

B.Tech.: Artificial Intelligence and Data Science

5AID5-11: Fundamentals of Blockchain

Credit: 2
2L+0T+0P

Max. Marks: 100(IA:30, ETE:70)
End Term Exam: 3 Hours

Course Objectives:

1. The students should be able to understand a broad overview of the essential concepts of blockchain technology.
2. To familiarize students with Bitcoin protocol followed by the Ethereum protocol – to lay the foundation necessary for developing applications and programming.
3. Students should be able to learn about different types of blockchain and consensus algorithms.

Course Outcomes: After completion of the course, students would be able to:

1. To explain the basic notion of distributed systems.
2. To use the working of an immutable distributed ledger and trust model that defines blockchain.
3. To illustrate the essential components of a blockchain platform.

Detailed Syllabus: (per session plan)

UNIT	Contents
1	Basics: The Double-Spend Problem, Byzantine Generals' Computing Problems, Public-Key Cryptography, Hashing, Distributed Systems, Distributed Consensus.
2	Technology Stack: Blockchain, Protocol, Currency. Bitcoin Blockchain: Structure, Operations, Features, Consensus Model, Incentive Model
3	Ethereum Blockchain: Smart Contracts, Ethereum Structure, Operations, Consensus Model, Incentive Model.
4	Tiers of Blockchain Technology: Blockchain 1.0, Blockchain 2.0, Blockchain 3.0, Types of Blockchain: Public Blockchain, Private Blockchain, Semi-Private Blockchain, Sidechains.
5	Types of Consensus Algorithms: Proof of Stake, Proof of Work, Delegated Proof of Stake, Proof Elapsed Time, Deposit-Based Consensus, Proof of Importance, Federated Consensus or Federated Byzantine Consensus, Practical Byzantine Fault Tolerance. Blockchain Use Case: Supply Chain Management.

TEXT BOOKS:

1. Kirankalyan Kulkarni, Essentials of Bitcoin and Blockchain, Packt Publishing.
2. Anshul Kaushik, Block Chain & Crypto Currencies, Khanna Publishing House.
3. Tiana Laurence, Blockchain for Dummies, 2nd Edition 2019, John Wiley & Sons.
4. Mastering Blockchain: Deeper insights into decentralization, cryptography, Bitcoin, and popular Blockchain frameworks by Imran Bashir, Packt Publishing (2017).

REFERENCE BOOKS:

1. Blockchain: Blueprint for a New Economy by Melanie Swan, Shroff Publisher O'Reilly Publisher Media; 1st edition (2015).
2. Mastering Bitcoin: Programming the Open Blockchain by Andreas Antonopoulos.

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RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Syllabus

B.Tech.: Artificial Intelligence and Data Science

5AID5-12: Probability and Statistics for Data Science

Credit: 2

Max. Marks: 100(IA:30, ETE:70)

2L+0T+0P

End Term Exam: 3 Hours

Objectives:

1. To provide advanced statistical background for analysing data and drawing inferences from that analysis.
2. Predicative Analytics using liner and generalized liner model.

Outcomes:

After completion of the course, students would be able to:

- Students will be able to learn advanced statistical technique and apply them to the analysis of real data sets from different fields.

Detailed Syllabus: (per session plan)

UNIT	Contents
1	Descriptive Statistics: a) Measures of Central Tendencies – Grouped and Ungrouped Data; Mean, Sample Mean– Weighted mean; Median, Quartiles, b) Deciles, and Percentiles, Box plot, Mode Measures of Variability– Dispersion, Range, Standard deviation, Population v/s sample variance and standard deviation, Skewness, Kurtosis.
2	Introduction to Probability and Sampling distribution: a) Methods of Assigning probabilities, Probability Space, conditions of probability model, Events, simple and compound, Laws of probability, Probability density function, Cumulative distribution function, Expected values of Mean and Variance. Marginal, union, joint and conditional probabilities, Bayes' Theorem b) Random variables, discrete and continuous distributions, Expectation, moments of a distribution, Binomial, Poisson, uniform, and normal distributions, Normal approximation to the binomial distribution, Distributions of several random variables, moments of joint distributions, independence, covariance, correlation coefficient, Central Limit Theorem.
3	Hypothesis Testing: a) Large Sample estimation of the population parameters and Hypothesis testing: Basics of Estimating the populations mean and difference; estimating the proportion and difference; large sample test for population mean, difference; large sample test for proportion, difference. b) Estimation of a population variance: Sampling distribution of variance, estimation. c) Inferences from small sample: Student's t distribution; Small sample t test for following – A population mean, A difference between two means, Confidence interval.
4	Regression Model: a) least squares and linear regression: Introduction; Notation; Ordinary least squares; Regression to the mean; Linear regression; Residuals; Regression inference b) Multivariable regression: Multivariate regression; Multivariate examples; Adjustment; Residual variation and diagnostics; Multiple variables , Interaction Terms, Non-linear Transformations of the Predictors, Qualitative Predictors.
5	Generalized linear models: Logistic Regression, Binary outcomes, Count outcomes, Multiple Logistic Regression ANOVA/MANOVA: Chi-Square and Analysis of Variance, Multivariate analysis of variance Extension of regression analysis: Ridge Regression, The Lasso



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Text Books:

1. An Introduction to Statistical learning with application in R . Hastie T, Robert T. (2014).
Springer Science Business Media: New York

Reference Books:

1. Statistics for Management, Seventh Edition, by Richard I. Levin, David S. Rubin, Pearson
2. An Introduction to Categorical Data Analysis. Agresti, A. (2012). John Wiley & sons
3. The Element of Statistical Learning, Data mining, Inference and Prediction. Hastie, T, Tibshirani, R, & Friedman, J. (2011). New York: Springer Series in Statistics.
4. Hair, Black, Babin, Anderson and Tatham (2009). Multivariate Data Analysis, Pearson



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B.Tech.: Artificial Intelligence and Data Science

5AID5-13: Programming for Data Science

Credit: 2
2L+0T+0P

Max. Marks: 100(IA:30, ETE:70)
End Term Exam: 3 Hours

Course Objectives:		
1. To provide necessary knowledge on data manipulation and to perform analysis on the practical problems using statistical and machine learning approach		
2. To generate report and visualize the results in graphical form using programming tool		
Expected Course Outcome:		
1. Ability to gain basic knowledge on data science		
2. Convert the real time data into suitable form for analysis		
3. Gain the insights from the data through statistical inferences		
4. Develop suitable models using machine learning techniques and to analyze its performance		
5. Identify the requirement and visualize the results		
6. Analyze on the performance of the model and the quality of the results		
Unit:1	INTRODUCTION	4 hours
Data Science: Introduction to Data Science – Digital Universe – Sources of Data – Information Commons – Data Science Project Life Cycle: OSEMN Framework		
Unit:2	DATA PREPROCESSING & CONCEPT LEARNING	6 hours
Introduction to Data Preprocessing – Reading, Selecting, Filtering Data – Filtering Missing Values – Manipulating, Sorting, Grouping, Rearranging, Ranking Data Formulation of Hypothesis – Probabilistic Approximately Correct Learning - VC Dimension – Hypothesis elimination – Candidate Elimination Algorithm		
Unit:3	ESSENTIALS OF R	8 hours
R Basics - data types and objects - control structures – data frame -Feature Engineering - scaling, Label Encoding and One Hot Encoding, Reduction		
Unit:4	MODEL FIT USING R	8 hours
Regression Models- Linear and Logistic Model, Classification Models – Decision Tree, Naïve Bayes, SVM and Random Forest, Clustering Models – K Means and Hierarchical clustering		
Unit:5	VISUALIZATION	6 hours
Data visualization: Box plot, histogram, scatter plot, heat map – Working with Tableau – Outlier detection – Data Balancing		
Unit:6	PERFORMANCE EVALUATION in R	4 hours
Loss Function and Error: Mean Squared Error, Root Mean Squared Error – Model Selection and Evaluation criteria: Accuracy, Precision, F1 score, Recall Score – Binary Predictive Classification – Sensitivity – Specificity.		

Text Book(s)

1. Ethem Alpaydin, Introduction to Machine Learning, Fourth Edition, MIT Press, 2020
2. Hadley Wickham, Garrett Grolemund, R for data science : Import, Tidy, Transform, Visualize, And Model Data Paperback, 2017

Reference Books

1. Han, J., Kamber, M., Pei, J. Data mining concepts and techniques. Morgan Kaufmann. 2011
2. Carl Shan, Henry Wang, William Chen, Max Song. The Data Science Handbook: Advice and Insight from 25 Amazing Data Scientists. The Data Science Bookshelf. 2016
3. James, G., Witten, D., T., Tibshirani, R. An Introduction to statistical learning with applications in R. Springer. 2013



RAJASTHAN TECHNICAL UNIVERSITY, KOTA

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B.Tech.: Artificial Intelligence and Data Science

5AID4-21: Computer Graphics & Multimedia Lab

Credit: 1
0L+0T+2P

Max. Marks:100 (IA:60, ETE:40)
End Term Exam: 2 Hours

SN	List of Experiments
1	Implementation of Line, Circle and ellipse attributes
2	To plot a point (pixel) on the screen
3	To draw a straight line using DDA Algorithm
4	Implementation of mid-point circle generating Algorithm
5	Implementation of ellipse generating Algorithm
6	Two Dimensional transformations - Translation, Rotation, Scaling, Reflection, Shear
7	Composite 2D Transformations
8	Cohen Sutherland 2D line clipping and Windowing
9	Sutherland – Hodgeman Polygon clipping Algorithm
10	Three dimensional transformations - Translation, Rotation, Scaling
11	Composite 3D transformations
12	Drawing three dimensional objects and Scenes
13	Generating Fractal images



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B.Tech.: Artificial Intelligence and Data Science

5AID4-22: Compiler Design Lab

Credit: 1
0L+0T+2P

Max. Marks: 100 (IA:60, ETE:40)
End Term Exam: 2 Hours

SN	List of Experiments
1	Introduction: Objective, scope and outcome of the course.
2	To identify whether given string is keyword or not.
3	Count total no. of keywords in a file. [Taking file from user]
4	Count total no of operators in a file. [Taking file from user]
5	Count total occurrence of each character in a given file. [Taking file from user]
6	Write a C program to insert, delete and display the entries in Symbol Table.
7	Write a LEX program to identify following: 1. Valid mobile number 2. Valid url 3. Valid identifier 4. Valid date (dd/mm/yyyy) 5. Valid time (hh:mm:ss)
8	Write a lex program to count blank spaces, words, lines in a given file.
9	Write a lex program to count the no. of vowels and consonants in a C file.
10	Write a YACC program to recognize strings aaab, abbb using a^nb^n , where $b \geq 0$.
11	Write a YACC program to evaluate an arithmetic expression involving operators +, -, * and /.
12	Write a YACC program to check validity of a strings abcd, aabbcd using grammar $a^nb^nc^md^m$, where $n, m > 0$
13	Write a C program to find first of any grammar.



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B.Tech.: Artificial Intelligence and Data Science

5AID4-23: Analysis of Algorithms Lab

Credit: 1
OL+0T+2P

Max. Marks: 100 (IA:60, ETE:40)
End Term Exam: 2 Hours

SN	List of Experiments
1	Sort a given set of elements using the Quicksort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
2	Implement a parallelized Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
3	a. Obtain the Topological ordering of vertices in a given digraph. b. Compute the transitive closure of a given directed graph using Warshall's algorithm.
4	Implement 0/1 Knapsack problem using Dynamic Programming.
5	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.
6	Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.
7	a. Print all the nodes reachable from a given starting node in a digraph using BFS method. b. Check whether a given graph is connected or not using DFS method.
8.	Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.
9.	Implement All-Pairs Shortest Paths Problem using Floyd's algorithm.
10	Implement N Queen's problem using Back Tracking.



RAJASTHAN TECHNICAL UNIVERSITY, KOTA

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B.Tech.: Artificial Intelligence and Data Science

5AID4-24: Advance Java Lab

Credit: 1

Max. Marks: 100 (IA:60, ETE:40)

L+OT+2P

End Term Exam: 2 Hours

SN	List of Experiments
1	Introduction To Swing, MVC Architecture, Applets, Applications and Pluggable Look and Feel, Basic swing components : Text Fields, Buttons, Toggle Buttons, Checkboxes, and Radio Buttons
2	Java database Programming, java.sql Package, JDBC driver, Network Programming With java.net Package, Client and Server Programs, Content And Protocol Handlers
3	RMI architecture, RMI registry, Writing distributed application with RMI, Naming services, Naming And Directory Services, Overview of JNDI, Object serialization and Internationalization
4	J2EE architecture, Enterprise application concepts, n-tier application concepts, J2EE platform, HTTP protocol, web application, Web containers and Application servers
5	Server side programming with Java Servlet, HTTP and Servlet, Servlet API, life cycle, configuration and context, Request and Response objects, Session handling and event handling, Introduction to filters with writing simple filter application
6	JSP architecture, JSP page life cycle, JSP elements, Expression Language, Tag Extensions, Tag Extension API, Tag handlers, JSP Fragments, Tag Files, JSTL, Core Tag library, overview of XML Tag library, SQL Tag library and Functions Tag library

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